

Evaluating Integrals

Today's Goals

- ◆ You should be able to apply integral formulas for basic functions, and understand where they come from.
- ◆ You should be able to find antiderivatives of sums, differences, and constant multiples of functions.
- ◆ You should be able to verify that you've correctly found an antiderivative of a function by taking a derivative.

With the Fundamental Theorem of Calculus and our knowledge of derivatives, we can figure out some useful formulas for integration:

Basic Indefinite Integrals

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| ◆ When $n \neq -1$ is a constant, $\int x^n dx = \underline{\hspace{2cm}}$ | ◆ $\int \cos x dx = \underline{\hspace{2cm}}$ |
| ◆ $\int \frac{1}{x} dx = \underline{\hspace{2cm}}$ | ◆ $\int \sec^2 x dx = \underline{\hspace{2cm}}$ |
| ◆ $\int e^x dx = \underline{\hspace{2cm}}$ | ◆ $\int \sec x \tan x dx = \underline{\hspace{2cm}}$ |
| ◆ When $b > 0$ is a constant, $\int b^x dx = \underline{\hspace{2cm}}$ | ◆ $\int \frac{1}{1+x^2} dx = \underline{\hspace{2cm}}$ |
| ◆ $\int \sin x dx = \underline{\hspace{2cm}}$ | ◆ $\int \frac{1}{\sqrt{1-x^2}} dx = \underline{\hspace{2cm}}$ |

1. Evaluate the following definite and indefinite integrals.

(a) $\int \frac{5}{7x} dx$

(b) $\int_{-e}^{-1/e} \frac{5}{7x} dx$

$$(c) \int (e^\pi + ey + \pi^y) dy$$

$$(d) \int (\pi + x)\sqrt{x} dx$$

$$(e) \int \left(\frac{e}{x^2} + 2^x \cdot 3^x \right) dx$$

$$(f) \int_{-3}^0 \left(\sqrt{9 - x^2} + x^2 \right) dx$$

2. A busy bagel shop starts making bagels at 5 am each morning. One day, the rate at which bagels are made is

$$r(t) = \begin{cases} 30\sqrt{2t} & 0 \leq t \leq 8 \\ 30(12 - t) & 8 < t \leq 12 \end{cases} \text{ bagels per hour,}$$

where t is measured in hours after 5am. The bagel shop is open from 7 am–5 pm; on this day, customers come at a steady rate of 60 customers per hour between 7 am and 5 pm. Each customer orders one bagel and receives it immediately.

- (a) At what times on this day is the number of bagels the shop has available increasing? At what times is it decreasing?

- (b) At the end of the day, any unsold bagels are donated to a local food pantry. How many bagels are donated at the end of this day?

- (c) What is the largest number of bagels the shop has available at any point during this day?

- (d) Are there times during the day when the shop has exactly 310 bagels available? If so, at how many times?

3. True or false:

(a) If F is an antiderivative of f and G is an antiderivative of g , then $F + G$ is an antiderivative of $f + g$.

(b) If F is an antiderivative of f and G is an antiderivative of g , then $F \cdot G$ is an antiderivative of $f \cdot g$.

4. (a) Decide whether each of the following definite integrals is positive or negative.

i. $\int_{2\pi/3}^{3\pi/4} \frac{\cos^2 t - 1}{\cos^2 t} dt$

ii. $\int_0^{1/2} \left(\sqrt{\frac{1}{4} - x^2} + \frac{1}{\sqrt{1 - x^2}} \right) dx$

(b) Now calculate the integrals, simplifying your answers.